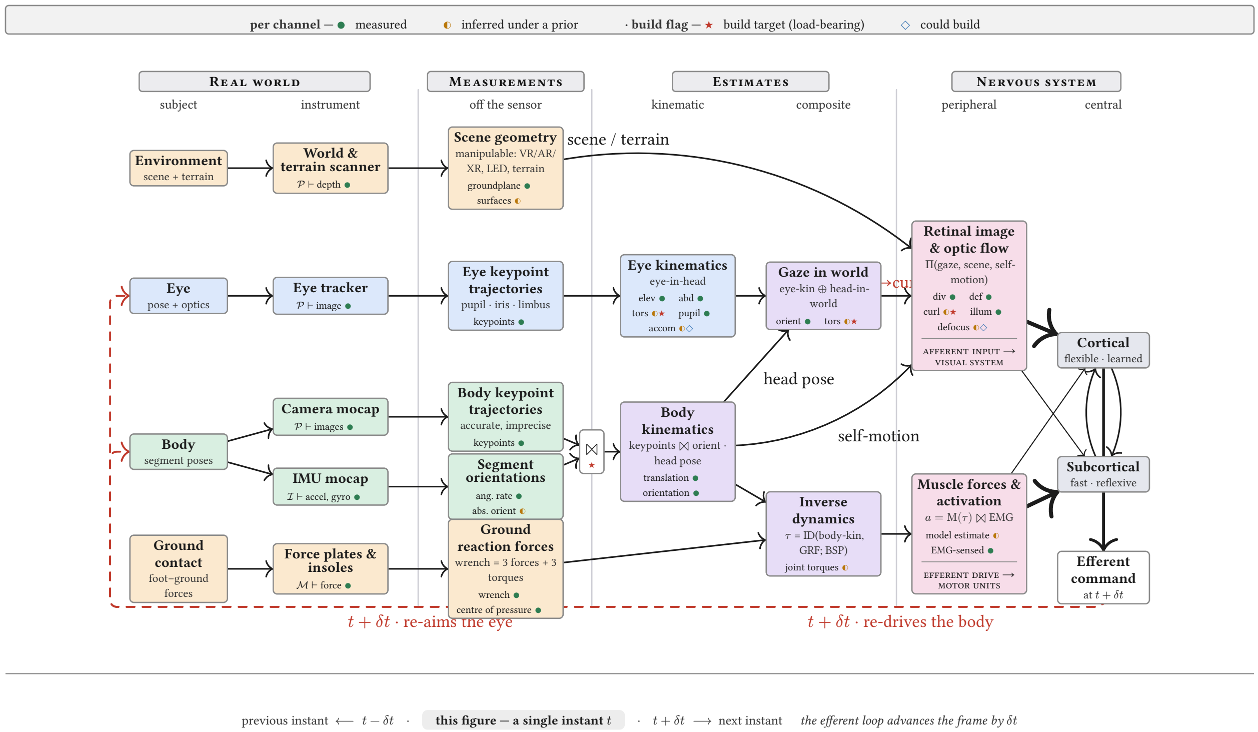

Reading the chart as a calculus each value is a bundle of **channels** (degrees of freedom); the top legend grades each channel.

Notation

$\mathcal{E}$	an **energy** a sensor is sensitive to: $\mathcal{P}$ photons · $\mathcal{I}$ inertial · $\mathcal{M}$ contact
$\mathcal{E} \vdash s$	**transduce** — a sensor **yields** ($\vdash$) a raw **signal** s from an energy (image · inertial · force)
$\to$	**derive** — a signal is processed into a measurement, then an estimate
$\oplus$	**compose** — chain relative poses; the shared reference frame cancels
$\bowtie$	**fuse** — union the measured subspaces of two estimators
Π	**project** — the only operator that maps input DOF onto output channels
$\approx v^\star$	**approximates** the true latent value $v^\star$

The chains — energy $\vdash$ signal $\to$ measurement $\to$ estimate

Eye	$\mathcal{P} \vdash$ image $\to$ eye keypoints $\to$ eye-kin (elev, abd, tors, pupil, accom)
Body	$\mathcal{P} \vdash$ images $\to$ body keypoints; $\mathcal{I} \vdash$ accel+gyro $\to$ segment orient; body-kin = keypoints $\bowtie$ orient
Gaze	g = eye-kin $\oplus$ head-in-world

Into the nervous system

Retinal input	$\rho = \Pi(g, \text{scene}, \dot{x}) \approx \rho^\star$ retinal image & optic flow (afferent)
Kinetics	$\mathcal{M} \vdash$ force $\to$ GRF; $\tau = \mathrm{ID}(\text{body-kin}, \text{GRF}; \text{BSP})$
Motor	$a = \mathrm{M}(\tau) \bowtie \mathrm{EMG} \approx a^\star$ muscle drive (efferent)

Figure 1: **The measurement chain of a Dome.** Reading across four bands — **real world ⟶ measurements ⟶ estimates ⟶ nervous system.** Sensors transduce physical energies into measurements (one step off the sensor); fusion ($\bowtie$) and pose composition ($\oplus$) build the kinematic and composite estimates that project (Π) onto the two points where the body meets the nervous system — the retinal stimulus entering vision and the muscle drive leaving the motor system. Every box is a bundle of **channels**; each channel is marked on two axes — measured (●) or inferred under a prior (◐) now, and, if unmeasured, a committed build target (★) or merely feasible (◇). The eye path makes the argument concrete: ocular torsion is inferred under Listing's law today and corrupts exactly the retinal **curl** channel — which is why it is a build target. The two peripheral signals project to the **central nervous system** (cortical + subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant t**: it is preceded by the infinitesimal before it and followed by the one after, and the dashed red loop is what advances the frame by δt. The calculus strip beneath the diagram gives the full reading rule, every symbol defined.